Expt. No. 8

Title: To design the layout of 6T SRAM using micro-wind.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

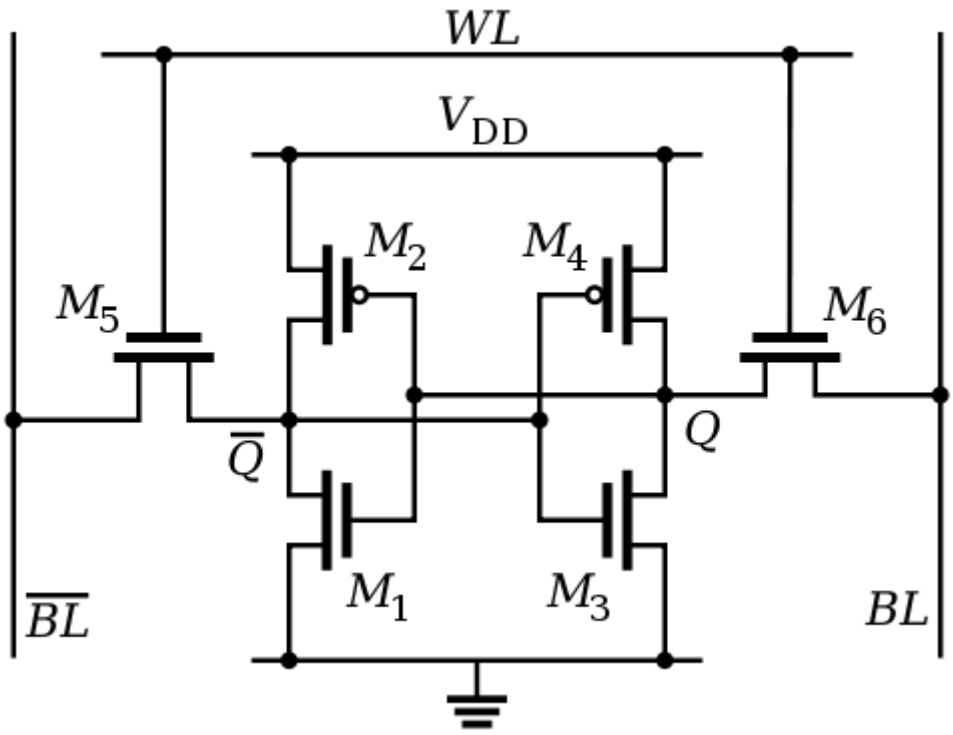
Experiment No. 8

Aim: To design the layout of 6T SRAM using micro-wind.

Software Required: Micro-wind software

Hardware: computer

Circuit Diagram:



Theory:

The memory cell shown here forms the basis for most static random-access memories in CMOS technology. It uses six transistors to store and access one bit. The four transistors in the center form two cross-coupled inverters. In actual devices, these transistors are made as small as possible to save chip-area, and are very weak. Due to the feedback structure, a low input value on the first inverter will generate a high value on the second inverter, which amplifies (and stores) the low value on the second inverter. Similarly, a high input value on the first inverter will generate a low input value on the second inverter, which feeds back the low input value onto the first inverter. Therefore, the two inverters will store their current logical value, whatever value that is.

The two lines between the inverters are connected to two separate bit-lines via two n-channel pass-transistors (left and right of the cell). The gates of those transistors are driven

by a word-line. In a larger SRAM, the word-line is used to address and enable all bits of one memory word (e.g. all 32 bits at address 0xcafe from a 64Kx32 SRAM chip). As long as the word-line is kept low, the SRAM cell is disconnected from the bit-lines. The inverters keep feeding themselves, and the SRAM stores its current value.

1) NWELL/PWELL width =6lambda
2) NWELL/PWELL spacing=6lambda
3) Active spacing =2lambda
4) Poly-width =2lambda
5) Poly-spacing=2lambda
6) Metal width =3lambda
7) Metal to metal width =3lambda
8) Mask=2lambda*2lambda

Procedure:

1] Start micro-wind software for layout**.**

2] Draw the 6T SRAM following the design.

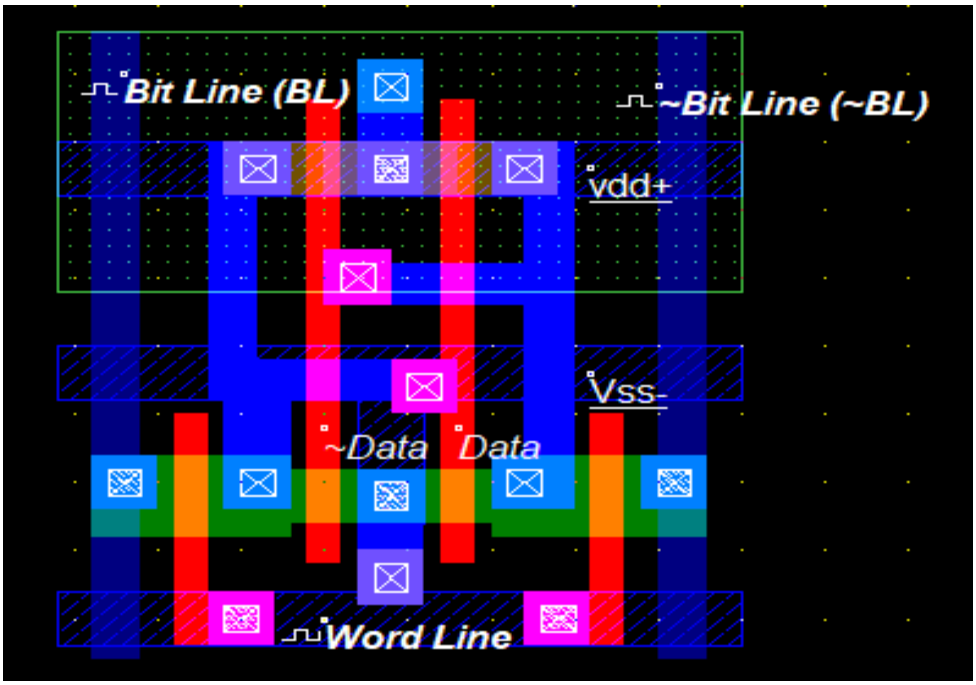
3] Click on DRC at every stage of design to avoid error.

4] Observe various output while using simulation output.

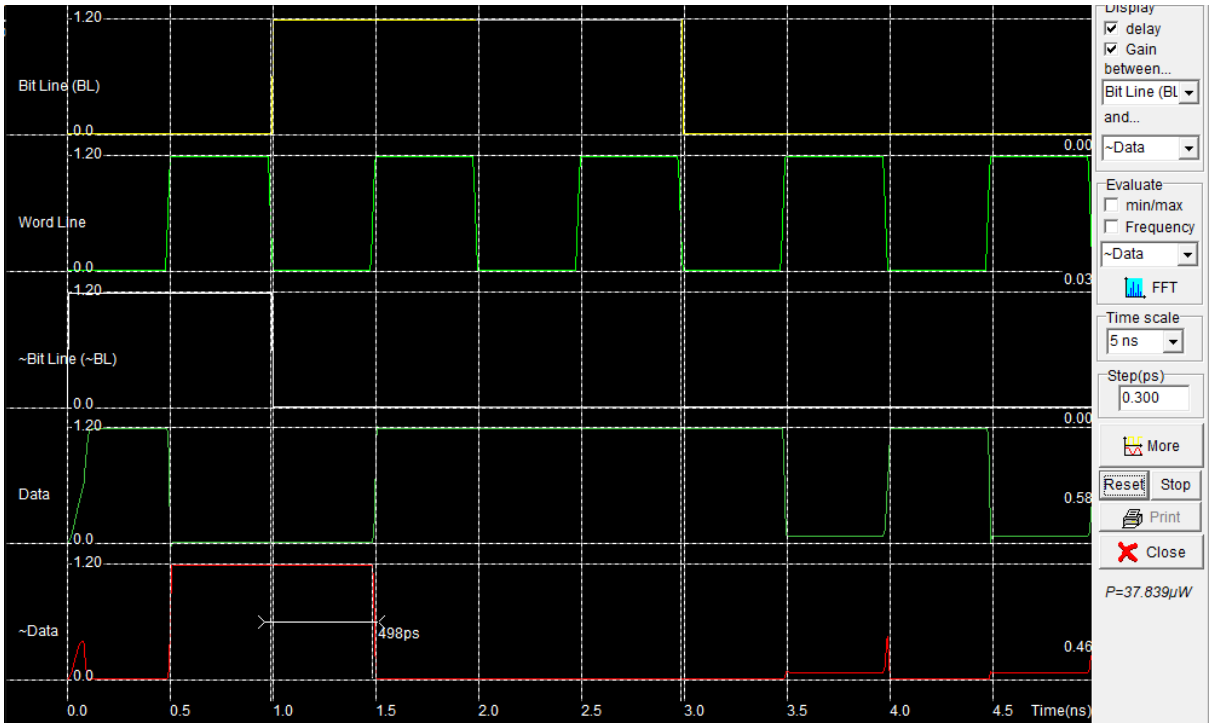
5] Software design and finally run a DRC.

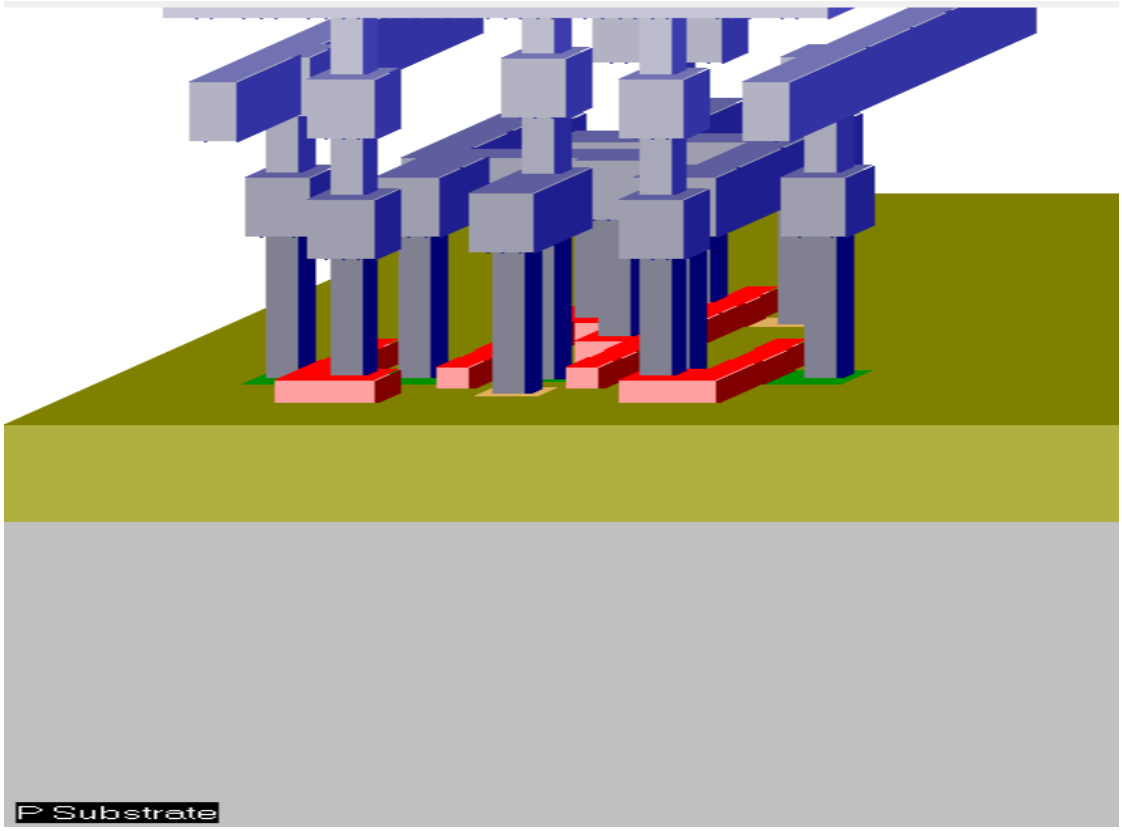
6] Record the output waveform observed.

Layout:



Output:


Bit Line (BL)
Word Line
~Bit Line (~BL)
Data
~Data
1.20
0.0
498ps
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Time(ns)
Display
delay
Gain
between...
Bit Line (BL
and...
~Data
Evaluate
min/max
Frequency
~Data
FFT
Time scale
5 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=37.839µW


P Substrate

Pre Lab-Questions:

1 Describe the various effects of scaling in 6T SRAM**?**
2 What are the limitations in increasing the power supply to reduce delay?

Post Lab Questions:

3 What's the critical path in a SRAM?
4 How does the size of PMOS Pull Up transistors (for bit & bit- lines) affect SRAM's performance?

Conclusion: Thus, we have designed the layout of the 6T SRAM using micro-wind

CO's covered:

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